

Uniqueness of the Minimal Finite-Cell Variational Reduction: A Derivation Target for the Fine-Structure-Scale Pressure Closure

Omar Iskandarani
Independent Researcher, Appingedam, The Netherlands

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Abstract

The companion papers isolate a dimensionless fine-structure-scale closure based on the ideal-trefoil ropelength, a spherical pressure cell, and a finite-cell BEM/NLS stationary point. The remaining question is whether the coefficients $N_p = 4$, $\sigma = 11/3$, $\chi_R = 2$, and $q_\phi = 1$ are merely closure choices or consequences of a more restrictive variational reduction. This note states and proves a conditional uniqueness theorem: under $SO(3)$ covariance, locality on the spherical cell boundary, positive quadratic order, volume and centre-displacement pressure constraints, inner/outer pressure reciprocity, and selection of the leading $1/r$ exterior phase field, the minimal finite-cell variational reduction uniquely gives

$$N_p = 4, \quad \sigma = \frac{11}{3}, \quad \chi_R = 2, \quad q_\phi = 1.$$

Consequently,

$$E_p^{\text{NLS}} = \frac{16\pi}{3} \mathcal{L}_K \left(1 - \frac{11}{48\mathcal{L}_K^2} \right)$$

is derived within the stated finite-cell variational reduction. The result is not claimed to be a complete first-principles derivation from the full microscopic field equations; rather, it identifies the precise reduction assumptions under which the pressure-cell closure is unique.

1 Purpose and status

The preceding companion papers separate three levels of claim. The first is an algebraic closure audit. The second is a spherical/NLS pressure-cell construction. The third is a finite-cell BEM/NLS stationary-point calculation and conditional far-field consistency test. The present note supplies the missing logical bridge between a free closure model and a derived effective model: it proves uniqueness of the minimal finite-cell variational reduction under explicitly stated axioms.

The status of the result is therefore

derived within the unique minimal finite-cell variational reduction.

It is not, by itself,

derived from the full microscopic dynamics without reduction assumptions.

That stronger statement would require deriving the axioms below directly from the underlying field theory.

2 Axioms of the minimal finite-cell reduction

Let the finite cell be represented by a spherical boundary S^2 , with outward unit normal $\mathbf{n}$, pressure field p , dimensionless shell thickness $\eta = a/R_{\text{cell}}$, cell-radius ratio $\chi_R = R_{\text{cell}}/L_K$, and exterior phase field u . The minimal quadratic reduction is assumed to satisfy:

- A1. $SO(3)$ covariance on the spherical boundary.
- A2. Locality at quadratic order on S^2 .
- A3. Positive quadratic energy for stable boundary perturbations.
- A4. Pressure constraints only for total compression and centre displacement:

$$\int_{S^2} p \, d\Omega = P_0, \quad \int_{S^2} p \mathbf{n} \, d\Omega = \mathbf{P}_1.$$

- A5. Isotropic NLS/GP shell-gradient response at second order.
- A6. Reciprocal inner/outer pressure balance at leading scale order.
- A7. Leading far-field coupling carried only by the exterior $1/r$ harmonic phase mode.

These assumptions define the class in which the uniqueness statement is made.

3 The finite-cell effective action

At quadratic order the effective action separates into

$$\mathcal{A}_{\text{eff}} = \mathcal{A}_p[p] + \mathcal{A}_{\text{shell}}[\eta] + \mathcal{A}_\chi(\chi_R) + \mathcal{A}_\phi[u].$$

The four pieces are fixed below by the axioms, up to an overall normalization that cancels in the dimensionless coefficient gates.

4 Theorem 1: constrained pressure minimization gives $N_p = 4$

Consider

$$\mathcal{A}_p[p] = \frac{1}{2} \int_{S^2} |\nabla_{S^2} p|^2 \, d\Omega$$

subject to

$$\int_{S^2} p \, d\Omega = P_0, \quad \int_{S^2} p \mathbf{n} \, d\Omega = \mathbf{P}_1.$$

Introducing Lagrange multipliers λ_0 and $\boldsymbol{\lambda}_1$, the Euler–Lagrange equation is

$$-\Delta_{S^2} p = \lambda_0 + \boldsymbol{\lambda}_1 \cdot \mathbf{n}.$$

The constant function belongs to $\mathcal{H}_0$, and the coordinate functions n_x, n_y, n_z span $\mathcal{H}_1$. Hence the minimal pressure subspace is

$$\mathcal{P}_{\text{min}} = \mathcal{H}_0 \oplus \mathcal{H}_1.$$

Since

$$\dim \mathcal{H}_\ell = 2\ell + 1,$$

one obtains

$$N_p = \dim \mathcal{H}_0 + \dim \mathcal{H}_1 = 1 + 3 = 4.$$

This proves uniqueness of $N_p = 4$ inside the lowest $SO(3)$ -closed pressure subspace satisfying the volume and centre-displacement constraints. Modes $\ell \geq 2$ represent quadrupolar and higher shape deformations and are not required by the constraints.

5 Theorem 2: isotropic NLS/GP shell variation gives $\sigma = 11/3$

Let

$$\eta = \frac{a}{R_{\text{cell}}}.$$

The volume Jacobian for a relative inward shell displacement is

$$\frac{V(R-a)}{V(R)} = (1-\eta)^3 = 1 - 3\eta + 3\eta^2 + O(\eta^3).$$

After the first-order displacement is absorbed into the equilibrium radius, the second-order volume contribution is

$$\sigma_{\text{vol}} = 3.$$

The isotropic NLS/GP gradient term contributes through the transverse projector

$$P_{\perp} = I - \hat{\mathbf{r}}\hat{\mathbf{r}}.$$

Using

$$\langle \hat{r}_i \hat{r}_j \rangle_{S^2} = \frac{1}{3} \delta_{ij},$$

one gets

$$\langle P_{\perp} \rangle_{S^2} = \frac{2}{3} I.$$

Equivalently,

$$\langle \sin^2 \theta \rangle_{S^2} = \frac{\int_0^{\pi} \sin^2 \theta \sin \theta d\theta}{\int_0^{\pi} \sin \theta d\theta} = \frac{2}{3}.$$

Thus the second-order shell coefficient is

$$\sigma = \sigma_{\text{vol}} + \sigma_{\perp} = 3 + \frac{2}{3} = \frac{11}{3}.$$

6 Theorem 3: reciprocal pressure stationarity gives $\chi_R = 2$

Let

$$\chi_R = \frac{R_{\text{cell}}}{L_K}.$$

The leading outer pressure-extension cost is proportional to χ_R , while the inner crowding/compliance cost is reciprocal and proportional to the number of pressure sectors. Inner/outer reciprocity therefore fixes the minimal scale action to

$$A_{\chi}(\chi_R) = \chi_R + \frac{N_p}{\chi_R}.$$

Stationarity gives

$$\frac{dA_{\chi}}{d\chi_R} = 1 - \frac{N_p}{\chi_R^2} = 0,$$

so

$$\chi_R = \sqrt{N_p} = 2.$$

Equivalently, if

$$A_{\chi} = \chi_R + \frac{\lambda_{\chi}}{\chi_R},$$

then

$$\lambda_{\chi} = N_p = 4.$$

This is unique in the leading-order class of reciprocal, scale-homogeneous inner/outer pressure actions. If additional terms such as χ_R^2 , χ_R^{-2} , or $\log \chi_R$ are admitted at the same order, uniqueness is lost; such terms are excluded by the minimal scale-reciprocity axiom.

7 Theorem 4: the exterior phase Hessian fixes $q_\phi = 1$

An exterior harmonic field has the multipole expansion

$$u(r, \theta, \phi) = \sum_{\ell, m} a_{\ell m} \frac{Y_{\ell m}(\theta, \phi)}{r^{\ell+1}}.$$

Only $\ell = 0$ gives a $1/r$ field. The $\ell = 1$ sector begins at r^{-2} , and all $\ell \geq 2$ modes are shorter-ranged. Therefore the leading far-field charge is the constant boundary phase, i.e. the unique $H^0(S^2)$ mode. Since S^2 is connected,

$$q_\phi = \dim H^0(S^2) = 1.$$

For the exterior monopole

$$u(r) = \frac{\phi}{r},$$

one has

$$\int_{r \geq 1} |\nabla u|^2 d^3x = \int_1^\infty 4\pi r^2 \frac{\phi^2}{r^4} dr = 4\pi\phi^2.$$

Thus

$$\Lambda_\phi = 4\pi\mathcal{K}_{\text{cell}}.$$

The unit H^0 one-cell phase Hessian gives

$$\Lambda_\phi = \frac{E_{\text{eff}}}{2},$$

and therefore

$$\mathcal{K}_{\text{cell}} = \frac{E_{\text{eff}}}{8\pi}.$$

8 Resulting pressure scale

Combining Theorems 1–3,

$$E_p^{(0)} = N_p \frac{4\pi}{3} \mathcal{L}_K = \frac{16\pi}{3} \mathcal{L}_K.$$

Moreover,

$$\eta_K = \frac{1}{2\chi_R \mathcal{L}_K} = \frac{1}{4\mathcal{L}_K}.$$

With $\sigma = 11/3$,

$$\sigma\eta_K^2 = \frac{11}{3} \frac{1}{16\mathcal{L}_K^2} = \frac{11}{48\mathcal{L}_K^2}.$$

Therefore

$$E_p^{\text{NLS}} = \frac{16\pi}{3} \mathcal{L}_K \left(1 - \frac{11}{48\mathcal{L}_K^2} \right).$$

Theorem 4 then gives the far-field stiffness relation

$$\mathcal{K}_{\text{cell}} = \frac{E_{\text{eff}}}{8\pi},$$

so that the leading two-cell interaction has coefficient

$$\alpha_{\text{cell}} = \frac{1}{4\pi\mathcal{K}_{\text{cell}}} = \frac{2}{E_{\text{eff}}}.$$

9 Falsification and robustness checks

The uniqueness theorem depends on the minimal axioms. The following checks test those axioms rather than the algebra:

1. include the $\ell = 2$ pressure sector and verify that it enters only as a higher-order shape correction;
2. replace $P_\perp$ by a weighted transverse projector and verify that isotropic NLS/GP reduction fixes the weight to unity;
3. replace $A_\chi = \chi_R + N_p/\chi_R$ by $a\chi_R + bN_p/\chi_R$ and verify that inner/outer reciprocity enforces $a = b$;
4. compute the one-cell phase Hessian directly from the full finite-cell operator and verify $\Lambda_\phi = E_{\text{eff}}/2$.

10 Conclusion

The four coefficient gates are not free once the minimal finite-cell variational reduction is imposed. The pressure constraints uniquely select $\mathcal{H}_0 \oplus \mathcal{H}_1$ and hence $N_p = 4$; the isotropic shell second variation gives $\sigma = 11/3$; reciprocal inner/outer pressure stationarity gives $\chi_R = 2$; and the exterior $1/r$ phase field gives $q_\phi = 1$ and $\mathcal{K}_{\text{cell}} = E_{\text{eff}}/(8\pi)$. The resulting fine-structure-scale pressure closure is therefore derived within this unique minimal variational reduction. The remaining task is to derive the reduction axioms themselves from the full microscopic field equations.

References

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